Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{HSEL}	OSC_IN input pin low-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	V_{SS}	-	0.3 × V_{DD}	V
$t_{w(HSEH)}$ $t_{w(HSEL)}$	OSC_IN high or low time	Digital mode (HSEYBYP = 1, HSEEXT = 1)	7	-	-	ns
$DuCy_{HSE}$	OSC_IN duty cycle	Digital mode (HSEYBYP = 1, HSEEXT = 1)	45	-	55	%
$V_{HSE\_ext\_PP}$	OSC_IN peak-to-peak amplitude	Analog mode (HSEBYP = 1, HSEEXT = 0)	0.2	-	2/3 V_{DD}	V
V_{HSE_ext}	OSC_IN input range		0	-	V_{DD}	
$t_{r(HSE)}$, $t_{f(HSE)}$	OSC_IN rise and fall time		0.05 / f_{ext_ext}	-	0.3 / f_{ext_ext}	ns

1. *There is no specified rise and fall time for a digital input signal, but the V_{HSEH} and V_{HSEL} conditions must be fulfilled .*
2. *The DC component of the signal must ensure that the signal peaks are located between V_{DD} and V_{SS}.*

Figure 14. AC timing diagram for high-speed external clock source (digital mode)

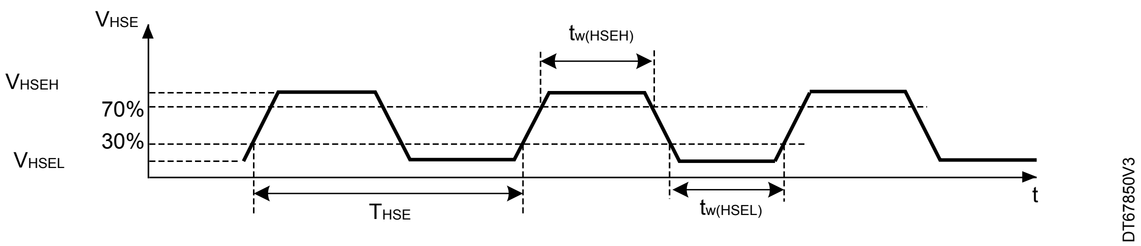

Figure 15. AC timing diagram for high-speed external clock source (analog mode)